

IT3040 – ITPM | Semester 2

EduConnect

Study Group Platform

An all-in-one MERN stack learning platform for university students

Group - WE_296_5.2

BSc (Hons) Information Technology | Year 3 | semester 2 | 11 Weeks

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Introduction

Problem Statement



Fragmented Resources

Students juggle multiple apps for notes, videos, quizzes, and discussions — breaking focus and wasting time.



No Peer Collaboration

No unified space to book Kuppi sessions, connect with tutors, or collaborate with classmates effectively.



No Progress Insight

Students lack visibility into their performance, strengths and weaknesses, making self-improvement difficult.



No Community Space

Lost items, found items, notices and campus events have no dedicated in-app board — students miss out.

Proposed Solution

EduConnect

A unified MERN-stack web platform purpose-built for university students, bringing together everything they need to learn, collaborate, and excel — in one place.

 Course & lecture resource management

 Mock exams, quizzes & instant grading

 Kuppi (peer study) session booking

 AI chatbot for 24/7 student support

 Payment gateway for premium content

 Real-time progress tracking dashboard



Student Community Board

Key Features



Course & Content Management

Browse, enroll & access lecture notes, PDFs & videos organized by week/topic/module.



Quiz & Mock Exam System

MCQ, True/False & short-answer quizzes with timer, auto-grading & answer review.



Kuppi Session Booking

Schedule and book peer study sessions; email confirmations & tutor notifications.



User Auth & Profiles

JWT-secured login, role-based access (Student / Tutor / Admin) & profile management.



AI Support Chatbot

OpenAI-powered 24/7 assistant giving contextual help and personalized study tips.



Payment Gateway

Stripe integration to unlock premium courses, quizzes & Kuppi sessions.

Technology Stack

MERN Stack

MongoDB · Express.js · React.js · Node.js

Frontend

React.js

Component library, Chart.js / Recharts, responsive design

Backend

Node.js + Express.js

REST APIs, JWT auth, Multer file uploads, bcrypt

Database

MongoDB

Schemas for users, courses, quizzes, bookings, payments

AI Integration

OpenAI GPT API

Natural language chatbot for student support

Payments

Stripe / PayPal

Secure checkout, webhooks, PCI compliance

File Storage

Multer

Lecture PDFs, video hosting, organized by course

Testing

Playwright

Automated UI testing for all key user journeys

Version Control

Git + GitHub

Feature branches, pull requests, meaningful commits

Responsibility Matrix

Member 1

Course & Content Management

Course catalog & enrollment system

Lecture notes & PDF/video upload

Content organization by week/topic

Ccommunity Board

Member 2

Quiz & Mock Exam System

Quiz creation & management

Quiz attempt UI with timer

Progress tracking dashboard

Member 3

Kuppi Booking & User Mgmt

Kuppi session scheduling & booking

JWT authentication & role-based access

User profile management

Member 4

AI Chatbot & Payment Gateway

AI-powered 24/7 support chatbot

Stripe/PayPal payment integration

Premium content unlocking system



Member 1 – Rathnayake R.M.N.P.D

Course & Content Management System

Course catalog · Lecture notes upload · Content organization



Inputs

- Student selects course & category
- Admin uploads PDFs & videos
- Student searches materials
- Student creates a community post
- Student selects post category



Outputs

- Courses displayed with details
- Enrollment confirmation
- Organized lecture folders
- Search results shown
- Recently accessed materials
- Post published on community feed
- Lost/Found notifications to all



Purpose

- Students can easily enroll
- All materials in one place
- Less time searching
- Better exam preparation
- Students share lost/found items
- Campus-wide notice board for all



Student Community Board

A university-wide notice board — post lost items, found items, events & campus notices



Lost Item

Post description & photo of something lost on campus. Others can reply if they find it.



Found Item

Found something on campus? Post it so the owner can claim it from you.



Announcement

Tutors & admins broadcast important university notices to all students.



Event

Promote clubs, sports events, study groups and social activities.



Help Request

Ask for academic help or volunteer to help peers with a subject.



Idea / Tip

Share a useful study tip, resource link or productivity idea.



Member 2 – Gunawardhana R.D.T.A

Quiz & Mock Exam System

Quiz creation · Quiz attempt interface · Progress tracking dashboard



Inputs



Admin creates quiz (MCQ, T/F)



Student attempts quiz



Student submits answers



Outputs



Automatic grading



Instant score & review



Performance analytics



Progress charts



Purpose



Real exam practice



Instant feedback



Track improvement over time



Tutors monitor performance



Member 3 – D.D Dharmappriya

Kuppi Session Booking & User Management

Kuppi scheduling · User authentication · Profile management



Inputs



Tutor creates session



Student books session



User registers or logs in



Outputs



Booking confirmation



Email notification sent



JWT authentication token



Role-based access enforced



Purpose



Easy group study booking



Secure & reliable login



Personalized user profiles



Member 4 – Gamage D.N

AI Chatbot & Payment Gateway

AI chatbot · Payment integration · Premium content unlocking



Inputs



Student asks chatbot a question



Student purchases premium quiz



Student enters payment details



Outputs



AI answer & study suggestions



Payment confirmation receipt



Premium content unlocked



Purpose



24/7 support without waiting



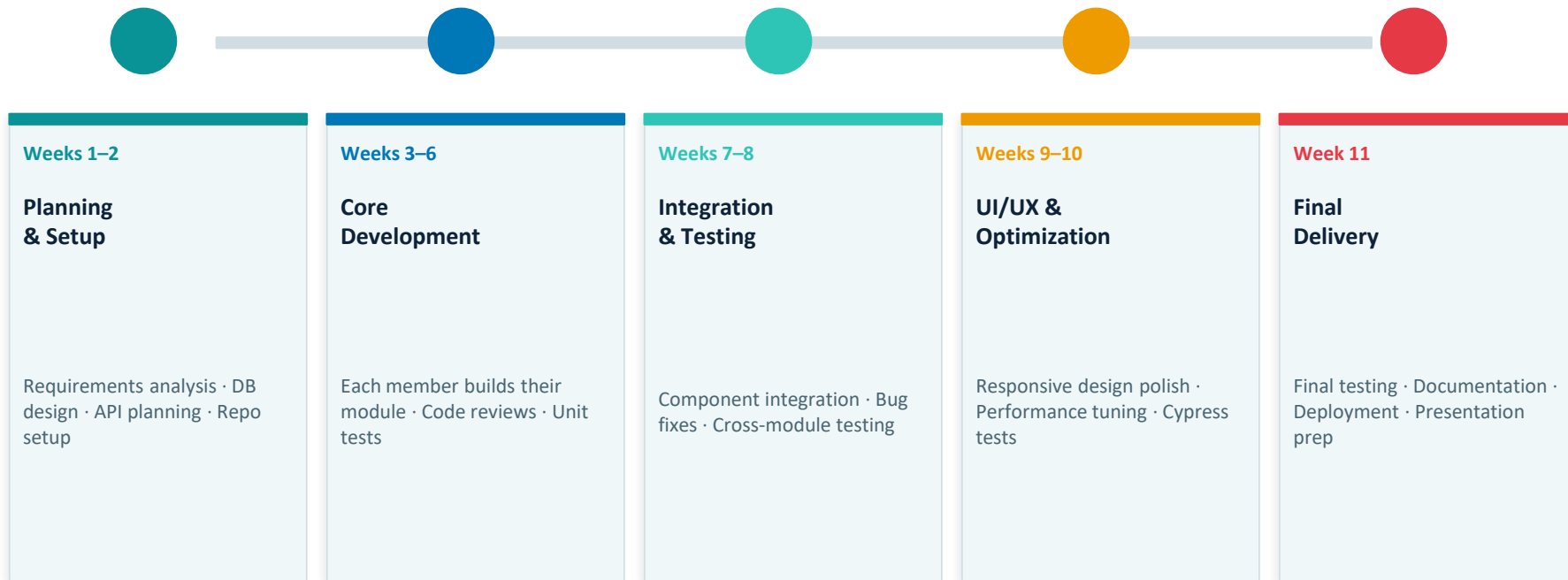
Monetization for creators



Secure transactions

Development Timeline

11-Week Project Schedule





Novelty of the Project

What makes EduConnect different from existing solutions



Integration

All-in-One Unified Platform

Unlike Moodle (LMS only) or Google Classroom, EduConnect merges courses, quizzes, Kuppi, AI chat, payments and a community board into one MERN-stack app — zero app-switching.



Monetisation

Integrated Freemium Model

Core content is free; premium quizzes and exclusive Kuppi sessions are gated behind Stripe/PayPal checkout — a sustainable monetisation model absent from open university LMS tools.



Social

University Community Board

A structured in-app notice board for lost/found items, campus events and help requests — absent from all mainstream LMS platforms and purpose-built for university social needs.



AI-Powered

Context-Aware AI Chatbot

Powered by OpenAI GPT, the bot gives study suggestions tailored to the student's enrolled modules and past quiz performance — not just generic answers.



Sri Lanka

Kuppi Peer-Study Booking

'Kuppi' is a uniquely Sri Lankan peer-tutoring tradition. EduConnect is the first platform to formalise scheduling with booking confirmation, email alerts and role-based access control.

Thank You

EduConnect — Study Group Platform

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Open for Questions